

STRENGTH TRAINING REFERENCE

# Mountain Strength

A working library and session templates for the road to the Gran Trail Costa Blanca 102K — built around descents, vert, and a left chain that needs respect.

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Target race  
**GTCB 102K · 6,280m+**

Start  
**Fri 13 Nov 2026, 23:00**

One session / week  
**Pairs with erg day**

# How to use this book

A finish-focused mountain ultra is won by legs that still brake on the descents at hour 18 and a body that stays upright and loaded for 24 hours. This strength work serves that — not aesthetics, not max numbers.

## READ THIS FIRST — THE NON-NEGOTIABLE

Alex Effer's prescribed corrective work is the **backbone** of your lower-leg and left-chain training. This book is a general strength supplement that sits *around* his rehab, never over the top of it. Where exercises here touch the calf or the left kinetic chain (foot → tibia → hip → knee → calf), **his guidance wins every time**. Run any new calf loading past him before adding volume — your working picture is plantarflexor hyperexcitability / EAMC, not a tissue you simply load harder.

## The principles

- **One quality session per week**, ~45–55 min including warm-up, on your erg/rowing day. Rowing is general aerobic and posterior-biased, so strength rounds out what the erg misses: single-leg control, eccentric braking, lateral chain, trunk endurance.
- **Rotate the emphasis** week to week (Templates 1 → 2 → 3 → repeat). Because it's only once weekly, each session is full-body with a different lean, not a body-part split.
- **Eccentric / descent-protective work is the priority bucket**. If a session gets cut short, keep those movements.
- **Quality over load**. Leave 2–3 reps in reserve (RIR). You're building durable tissue and control, not chasing a 1RM. Sore legs must not compromise the Friday Peak day.
- **Periodise down into the race**. Heaviest in base (Jun–Aug), trimmed through build, maintenance-only from ~mid-Oct so it never blunts your peak mountain weeks.
- **Never stack heavy lower-body the day before a long Peak day**. Keep strength early-week.

## Reading the prescriptions

| Notation      | Meaning                                                                                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 × 8 / leg   | 3 sets of 8 repetitions on <i>each</i> leg                                                                                                                                                 |
| Tempo 3-1-1-0 | Seconds for each phase: <b>lowering (eccentric)</b> – pause at bottom – lifting – pause at top. A high first number = slow, controlled lowering, which is the descent-protective stimulus. |
| RIR 2–3       | Reps In Reserve — stop with 2–3 good reps left in the tank. Never grind to failure.                                                                                                        |
| Iso :45       | Isometric hold for 45 seconds                                                                                                                                                              |

## Warm-up (every session, ~6–8 min)

- 2–3 min easy on the erg you just finished, or a brisk walk, to stay warm.
- Effer's activation/breathing drills if prescribed for that day — these come **first**.
- Glute activation: banded clamshells + glute bridges, ~15 reps each, to set the **glute → hamstring → calf** firing order before loading.

- 5-8 bodyweight reps of the session's first main lift as a primer set.

# Exercise library

Reference for every movement used in the templates. Tags show the primary purpose; doses are **starting points** for the base phase — progress per the rules at the back.

## A • Descent-protective & eccentric priority bucket

### Eccentric split squat

A1

PRIORITY

SINGLE-LEG

*The single best gym mimic of descending: loaded knee flexion under a slow, controlled lower. Builds quads' braking capacity so they survive 6,280m of downhill.*

**Setup:** Split stance, rear foot on the floor (or lightly on a low bench). Torso tall, weight through the front mid-foot.

**Execution:** Lower slowly and under control, front knee tracking over toes, until the front thigh is near parallel. Drive up at normal speed. The **lowering** is the whole point.

#### TEMPO

4-1-1-0

#### DOSE

3 × 6-8 / leg

#### PROGRESS

Hold dumbbells → slower eccentric

**Watch:** Keep the knee stable, no inward collapse — that's your left-chain pattern showing up. Stop the set if control goes.

### Step-down (forward & lateral)

A2

PRIORITY

SINGLE-LEG

*Directly trains the controlled, single-leg lowering of a downhill foot-plant, plus the hip/knee control that stops the left knee hyperextending and the foot over-pronating.*

**Setup:** Stand on a step/box (start ~15 cm, build height). One leg working, other leg hanging off the edge.

**Execution:** Slowly lower the free heel toward the floor by bending the standing leg — tap lightly, don't dump weight onto it — then control back up. **Lateral version:** step off to the side to load the hip abductors / lateral chain.

#### TEMPO

3-1-1-0

#### DOSE

3 × 8 / leg

#### PROGRESS

Higher box → light DB

**Watch:** Hips level (no drop on the free side), knee over mid-foot. This is a control drill before it's a strength drill.

## Spanish squat (isometric)

A3

### PRIORITY

*Knee-friendly quad loading with low joint stress — builds tendon-tolerant quad strength without the impact, useful when legs are already loaded from hill days.*

**Setup:** Loop a stout band around a fixed post and behind both knees. Lean back into the band, feet planted.

**Execution:** Sit back into a squat (~60–90° knee bend) with shins vertical and hold. Quads do the work; the band takes the shear.

#### HOLD

Iso :30–:45

#### DOSE

4–5 holds

#### PROGRESS

Add held weight / longer hold

## Reverse Nordic

A4

### PRIORITY

*Loaded eccentric lengthening of the quads through a big range — potent protection against late-race downhill quad blow-up. Advanced; ease in.*

**Setup:** Kneel tall on a pad, knees hip-width, torso and hips in one line.

**Execution:** Keeping a straight line from knee to head, lean back slowly under control as far as you can hold, then return. Small range first; build over weeks.

#### TEMPO

4-0-2-0

#### DOSE

2–3 × 5–8

#### PROGRESS

Greater lean angle

**Watch:** No hinging at the hip (don't fold) and no lower-back arch. Quads only. Start tiny — this gets sore.

## Eccentric heel drop

A5

### CALF · CHECK EFFER

*Classic calf/Achilles eccentric loading. Relevant to descending and push-off — but this is exactly the tissue under your working diagnosis.*

#### EFFER GOVERNS THIS ONE

Do not add eccentric calf loading on your own initiative. Use only the heel-drop volume, range and frequency Effer has set, and the cues he's given (heel elevation, big-toe wedge, breathing). The notes below are descriptive, not a prescription.

**Execution (if cleared):** Rise on both feet, shift to the working leg, lower the heel slowly below a step. Soleus bias: same with the knee bent.

## B • Single-leg strength & stability

### Bulgarian split squat

B1

#### SINGLE-LEG

*The workhorse single-leg strength builder. Mountain running is one leg at a time on uneven ground — this builds the unilateral strength and balance bilateral squats miss.*

**Setup:** Rear foot elevated on a bench, front foot far enough forward that the knee stays over mid-foot at the bottom.

**Execution:** Lower under control until the front thigh is ~parallel, drive through the front heel/mid-foot to stand. Keep the torso tall-ish.

#### TEMPO

3-0-1-0

#### DOSE

3 × 8-10 / leg

#### PROGRESS

DBs → goblet → slow tempo

**Watch left side:** hip stays level, knee doesn't dive in, no rotation. Quality over load on the weaker chain.

### High step-up

B2

#### SINGLE-LEG

*The closest gym mirror of power-hiking a steep climb — the GTCB's bread and butter. Trains hip/knee extension from a deep, loaded position.*

**Setup:** Box at mid-thigh height (knee bent ~90° when the foot's on it).

**Execution:** Drive up through the top leg with minimal push-off from the floor leg — make the working leg do it. Control the lower back down (don't just drop).

#### TEMPO

2-0-2-0

#### DOSE

3 × 8 / leg

#### PROGRESS

Hold DBs / loaded pack

Loading with your race pack ties it straight to power-hiking specificity.

### Single-leg Romanian deadlift

B3

#### SINGLE-LEG

#### POSTERIOR

*Builds the glute/hamstring chain unilaterally and trains balance + hip control on one leg — exactly the stability you need on technical, fatigued descents.*

**Setup:** Stand on one leg, soft knee, weight in the opposite hand (or both).

**Execution:** Hinge at the hip, back flat, free leg extending behind as a counterweight; lower until you feel the hamstring load, return by squeezing the glute.

#### TEMPO

3-0-1-0

#### DOSE

3 × 8 / leg

#### PROGRESS

Heavier DB / KB

**Watch:** Hips square to the floor (resist the open-up rotation) — this directly trains against your left-chain pattern.

## C • Posterior chain & glutes

### Romanian deadlift (bilateral)

C1

#### POSTERIOR

*Strong hamstrings and glutes power the climbs and protect the knees on descents. The hip hinge also reinforces the posture you'll hold for 24 hours.*

**Execution:** Barbell or DBs; hinge at the hips with a flat back, bar close to the legs, lower to mid-shin or where the back stays neutral, drive hips forward to stand.

#### TEMPO

3-0-1-0

#### DOSE

3 × 8-10

#### PROGRESS

Add load, keep form

### Hip thrust / glute bridge

C2

#### POSTERIOR

*Direct glute strength — the prime mover for climbing and the muscle that should fire first in your chain. Strong glutes take load off the over-eager calf.*

**Execution:** Upper back on a bench (thrust) or floor (bridge), barbell/DB across the hips, drive through heels to full hip extension, squeeze glutes hard at the top, lower with control.

#### TEMPO

2-1-1-0

#### DOSE

3 × 10-12

#### PROGRESS

Load → single-leg

### Banded lateral walk / clamshell

C3

#### POSTERIOR

#### LATERAL CHAIN

*Targets glute medius — the hip stabiliser that keeps your pelvis level and stops the left-side collapse pattern on every single foot-plant over 102km.*

**Execution:** Band above knees or ankles. **Lateral walk:** half-squat, step sideways keeping tension, hips level. **Clamshell:** side-lying, knees bent, open the top knee without rolling the pelvis back.

#### DOSE (WALK)

3 × 10 steps / way

#### DOSE (CLAM)

3 × 12-15 / side

#### PROGRESS

Stronger band

Doubles as warm-up activation. Useful most sessions.

## D • Trunk & postural endurance

A 24-hour race is a posture-endurance event. These build a trunk that holds you upright and stable when everything's tired — anti-movement > crunches.

### Suitcase carry (loaded carry)

D1

#### TRUNK

*Anti-lateral-flexion under load while walking — almost perfect transfer to staying tall and stable under a pack for hours. Underrated for ultras.*

**Execution:** Heavy DB/KB in one hand, walk tall and level, resisting the lean toward the weight. Switch sides.

#### DOSE

3 × ~30m / side

#### CUE

Ribs down, hips level

#### PROGRESS

Heavier / longer

### Pallof press

D2

#### TRUNK

*Anti-rotation — trains the trunk to resist the twist that creeps in with fatigue and feeds compensatory chains. Good carryover to your left-side rotation tendency.*

**Execution:** Band/cable at chest height to the side. Press the handle straight out and resist being rotated, hold, return. Stay square.

#### DOSE

3 × 10 / side

#### CUE

Slow, no twist

#### PROGRESS

Stronger band / longer hold

### Side plank (+ variations)

D3

#### TRUNK

#### LATERAL CHAIN

*Lateral trunk + glute medius endurance in one. Holds the pelvis stable late in the race when the left side wants to give.*

**Execution:** Forearm under shoulder, body in a line, hips up. Build from knees → full → top-leg raises. Hold steady, breathe.

#### DOSE

3 × :30–:45 / side

#### PROGRESS

Longer hold / leg raise

## Dead bug & bird dog

D4

### TRUNK

Trunk control with limb movement and breathing — fits naturally with Effer's breathing-led approach. Teaches a stable core while arms and legs work, which is what running actually is.

**Dead bug:** on your back, low back gently pressed down, slowly lower opposite arm + leg, exhale, return.

**Bird dog:** on hands and knees, extend opposite arm + leg without tipping, hold, switch.

#### DOSE

2–3 × 8 / side

#### CUE

Slow + exhale on reach

#### PROGRESS

Slower / longer levers

## E · Calf & lower leg

Effer-led zone

### WHOLE SECTION GOVERNED BY EFFER

Your calf is the limiter and the working diagnosis (plantarflexor hyperexcitability / EAMC on the altered-neuromuscular-control model) means **more calf load is not automatically better**. Treat this section as a checklist of what calf work *exists*, not licence to program it yourself. Volume, progression and whether to load at all are Effer's calls. Build strength without tipping into the over-firing pattern — and respect your own flagged tendency to over-test.

## Seated calf raise (soleus)

E1

### CALF · CHECK EFFER

*Soleus is the endurance calf muscle and a big player in distance running and descending control. Bent-knee position biases it.*

**Execution:** Seated, knees bent ~90°, weight on thighs, raise through the balls of the feet, full controlled lower. Heel-elevation and big-toe cues per Effer.

Only at the dose Effer sets.

## Standing calf raise (gastrocnemius)

E2

### CALF · CHECK EFFER

*Straight-knee calf raise biases the gastroc — the power calf for push-off. Single-leg builds the side-to-side balance running demands.*

**Execution:** Standing, knees straight, rise high, lower slowly with control. Progress bilateral → single-leg only when cleared.

Only at the dose Effer sets. Stop at any sign of the cramp/over-firing pattern.

# Session templates

Five ready-to-run sessions for your weekly erg-day slot. Rotate 1 → 2 → 3 across the weeks; use 4 when time-poor and 5 once you hit taper. Each assumes the 6–8 min warm-up first.

## Template 1 · Descent-Protective

~45 min · emphasis: eccentric quad / braking · base & build

| Exercise                 | Prescription            | Note                                    |
|--------------------------|-------------------------|-----------------------------------------|
| A1 Eccentric split squat | 3 × 6–8 / leg · 4-1-1-0 | Slow lower is the point                 |
| A2 Step-down (forward)   | 3 × 8 / leg · 3-1-1-0   | Hips level                              |
| A3 Spanish squat         | 4 × :30–:45 iso         | Or A4 Reverse Nordic 2×5 if progressing |
| C3 Banded lateral walk   | 3 × 10 steps / way      | Glute med                               |
| D3 Side plank            | 3 × :30–:45 / side      | Finisher                                |
| E1/E2 Calf               | Per Effer               | Only if prescribed                      |

## Template 2 · Single-Leg Strength & Climb

~50 min · emphasis: unilateral strength / power-hike · base & build

| Exercise                 | Prescription             | Note                           |
|--------------------------|--------------------------|--------------------------------|
| B2 High step-up          | 3 × 8 / leg · 2-0-2-0    | Load with pack for specificity |
| B1 Bulgarian split squat | 3 × 8–10 / leg · 3-0-1-0 | Knee over mid-foot             |
| B3 Single-leg RDL        | 3 × 8 / leg · 3-0-1-0    | Hips square                    |
| C2 Hip thrust            | 3 × 10–12 · 2-1-1-0      | Squeeze at top                 |
| D2 Pallof press          | 3 × 10 / side            | Anti-rotation                  |
| E1/E2 Calf               | Per Effer                | Only if prescribed             |

## Template 3 · Posterior Chain & Postural Endurance

~50 min · emphasis: glute/hamstring + trunk for the long haul · base & build

| Exercise               | Prescription       | Note                    |
|------------------------|--------------------|-------------------------|
| C1 Romanian deadlift   | 3 × 8–10 · 3-0-1-0 | Flat back, hinge        |
| B3 Single-leg RDL      | 3 × 8 / leg        | Balance + control       |
| C2 Hip thrust          | 3 × 10–12          | Glute drive             |
| D1 Suitcase carry      | 3 × ~30m / side    | Tall & level under load |
| D4 Dead bug / bird dog | 2–3 × 8 / side     | Breathe on the reach    |
| C3 Clamshell           | 3 × 12–15 / side   | Glute med finisher      |

## Template 4 · Full-Body Express

~30 min · for time-poor weeks — keeps the priorities ticking

| Exercise                                                                                            | Prescription   | Note              |
|-----------------------------------------------------------------------------------------------------|----------------|-------------------|
| B1 Bulgarian split squat                                                                            | 3 × 8 / leg    | Main strength     |
| A2 Step-down                                                                                        | 2 × 8 / leg    | Descent insurance |
| C2 Hip thrust                                                                                       | 2 × 12         | Glutes            |
| D1 Suitcase carry                                                                                   | 2 × 30m / side | Trunk             |
| If anything gets dropped further, keep the split squat and step-down — the descent-protective core. |                |                   |

## Template 5 · Taper Maintenance

~25–30 min · from ~mid-Oct (W17+) — maintain, don't fatigue

| Exercise                                                                                                              | Prescription           | Note                   |
|-----------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|
| B1 Bulgarian split squat                                                                                              | 2 × 6 / leg · moderate | RIR 3–4, no grind      |
| A2 Step-down                                                                                                          | 2 × 6 / leg            | Keep the pattern sharp |
| C3 Lateral walk + D3 side plank                                                                                       | 2 sets each            | Activation, light      |
| Cut load ~40–50%, keep movement quality and neural sharpness. Last real session ~7–10 days out, then only activation. |                        |                        |

# Fitting it to the 23-week build

One strength session a week, mapped to the phases of your run/hike/bike/row plan. The job of strength shifts from *building* to *maintaining* as the mountain volume takes over.

| Phase | Weeks            | Templates                         | Strength job                                                                                                                    |
|-------|------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Base  | W1-8 (Jun-Jul)   | 1 / 2 / 3 rotate                  | Build the durability bank. Heaviest loads, full sessions, progress weekly. Running's restricted, so strength is high-value now. |
| Build | W9-16 (Aug-Sep)  | 1 / 2 / 3, drop to 4 on big weeks | Maintain strength as hill/vert volume climbs. If a Peak B2B week is huge, use Template 4 so strength doesn't add fatigue.       |
| Peak  | W17-19 (Oct)     | 4, then 5                         | Pull volume right back. Maintenance only — protect the big mountain days.                                                       |
| Taper | W20-23 (Oct-Nov) | 5                                 | Keep the patterns alive, shed all fatigue. Last session ~7-10 days out.                                                         |

## Progression rules (base & build)

- **Add reps before load.** Hit the top of the rep range with 2-3 RIR for two sessions, then nudge weight up and drop back to the bottom of the range.
- **Slow the tempo** as a progression in its own right — a 5-second eccentric is harder and more descent-specific than just more weight.
- **Left side leads the dose.** On single-leg work, the weaker/affected side sets the reps and load; the strong side matches, not exceeds.
- **One variable at a time.** Don't add load, reps and range in the same week.

## When to back off (no heroics)

- Any hint of the calf cramp / over-firing pattern → stop calf work, tell Effer.
- Strength soreness bleeding into the Friday Peak day → cut volume or move the session earlier in the week.
- Left knee/hip control degrading mid-set → end the set; it's a quality drill, not an ego one.
- Big mountain B2B coming → downgrade to Template 4/5 that week. The race is built on the hills, not the gym.

### THE ONE-LINE SUMMARY

Strength here is insurance for the descents and the left chain, and an engine for the climbs — a supporting act to your hills, bike, row and Effer's rehab. Keep it consistent, keep it light enough to never cost you a Peak day, and taper it out so you arrive in Finestrat fresh.

